KS3 DIGITAL TECHNOLOGY

INTERNET OF THINGS AND SMART CITIES

Everyday objects that sense, think and act: how IoT devices use sensors, actuators and the cloud, how smart homes and smart cities work, why they can be risky, and how to design a responsible IoT solution.

LESSONS

1 Things that talk
2 Sensors and actuators
3 Smart homes
4 Smart cities
5 Hacking the things
6 Design an IoT solution

LESSON 1 OF 6

THINGS THAT TALK

The Internet of Things (IoT) is the network of everyday physical objects with sensors, processors and internet connections, so they can collect data, share it and act on it.

STARTER · 5 MINUTES

DO NOW

List everything in your home that is connected to the internet. Which would your grandparents never have imagined online?

TODAY

LEARNING INTENTIONS

WE ARE LEARNING TO

Explain what the Internet of Things is and describe the parts of an IoT system

SUCCESS CRITERIA: I CAN...

- ✓ I can explain what the Internet of Things is
- ✓ I can describe the four parts of an IoT system
- ✓ I can give examples of IoT in different areas of life

EVERYDAY THINGS, ONLINE

- The Internet of Things (IoT): physical objects connected to the internet
- They sense, share data and sometimes act on their own
- Doorbells, watches, thermostats, cars, farm sensors, bins, streetlights
- There are now well over 15 billion IoT devices worldwide

KEY WORDS

INTERNET OF THINGS	The network of everyday physical objects that have sensors, processors and internet connections.
SMART DEVICE	An everyday object that can collect data, connect to the internet and sometimes act automatically.
CLOUD	Remote servers, reached over the internet, that store and process data.
EMBEDDED SYSTEM	A small computer built into a larger device to carry out one specific job.

INSIDE AN IOT SYSTEM

- 1 Device: sensors and/or actuators, plus a small computer (embedded system)
- 2 Connectivity: Wi-Fi, Bluetooth, mobile (4G/5G) or low-power networks
- 3 Cloud: servers that store and analyse the data
- 4 App or dashboard: where people see data and control things

ACTIVITY · 15 MINUTES · PAIRS

IOT DETECTIVE

1. Look at the six product cards (e.g. a smart doorbell, a cattle collar, a bin sensor).
2. For each, identify its device, connectivity, cloud and app parts.
3. Decide which one would be most useful in Fermanagh and explain why.

In your workbook, answer:

? Describe the four parts of the smart doorbell.

? Which device would be most useful in Fermanagh, and why?

Your workbook: Lesson 1 · tick each step off as you go

IOT EVERYWHERE

- Home: smart speakers, thermostats, doorbells, plugs
- Health: fitness trackers, glucose monitors
- Farming: soil sensors, livestock trackers
- Industry and cities: machine monitoring, traffic sensors, smart bins

QUICK CHECK

1. What is the Internet of Things?

A A single website
B Everyday physical objects connected to the internet
C A type of cable
D A social network

2. What is an embedded system?

A A cloud server
B A computer game
C A small computer built into a device to do one job
D A type of cable

3. What does the cloud do in an IoT system?

A Stores and analyses data on remote servers
B Senses temperature
C Connects to Wi-Fi only
D Displays the app

4. Which is an example of an IoT device?

A A smart thermostat
B A pencil
C A paper calendar
D A wooden chair

QUICK CHECK: ANSWERS

1. **B: Everyday physical objects connected to the internet**
IoT connects physical things.

2. **C: A small computer built into a device to do one job**
Embedded systems are dedicated.

3. **A: Stores and analyses data on remote servers**
The cloud processes data.

4. **A: A smart thermostat**
It's connected and senses.

2 MARKS · IN YOUR WORKBOOK

EXIT TICKET

Explain what is meant by the Internet of Things.

Look back at today's success criteria: can you tick them all?

- ✓ I can explain what the Internet of Things is
- ✓ I can describe the four parts of an IoT system
- ✓ I can give examples of IoT in different areas of life

LESSON 2 OF 6

SENSORS AND ACTUATORS

Sensors measure the physical world and turn it into data (input).

STARTER · 5 MINUTES

DO NOW

A greenhouse waters its plants automatically when the soil is dry. What does it need to "know" and what does it need to "do"?

TODAY

LEARNING INTENTIONS

WE ARE LEARNING TO

Describe common sensors and actuators and how they form an input-process-output system

SUCCESS CRITERIA: I CAN...

- ✓ I can name common sensors and what they measure
- ✓ I can name common actuators and what they do
- ✓ I can describe an IoT system as input, process and output with feedback

SENSORS: THE SENSES

- Sensors measure something about the world and turn it into data
- Temperature, light, motion, moisture, sound, distance, air quality, location (GPS)
- Many are tiny and cheap
- A sensor alone just measures: it needs a processor to decide

KEY WORDS

SENSOR	A device that measures something in the physical world and turns it into data.
ACTUATOR	A device that makes something happen in the physical world, such as a motor, pump or light.
FEEDBACK LOOP	When a system's output affects its next input, letting it adjust automatically.

ACTUATORS: THE MUSCLES

- Actuators make something happen in the physical world
- Motors, pumps, valves, speakers, lights, locks, heaters
- Controlled by signals from the processor
- Example: open a window when the greenhouse is too hot

INPUT, PROCESS, OUTPUT AND FEEDBACK

- Input: sensor reading (soil moisture = 20%)
- Process: rule (IF moisture < 30% THEN water)
- Output: actuator (pump on for 10 seconds)
- Feedback: measure again and repeat, keeping conditions steady

ACTIVITY · 25 MINUTES · PAIRS

DESIGN A CONTROL SYSTEM

1. Choose a problem: a plant that needs watering, a room that gets too hot, or a pet bowl that needs refilling.
2. Choose the sensor(s) and actuator(s) you need.
3. Write the rule(s) as IF … THEN … statements.
4. Draw an input-process-output diagram with a feedback arrow.

In your workbook, answer:

? Write your sensor, actuator and rules.

? Explain how feedback keeps your system working without a person.

Your workbook: Lesson 2 · tick each step off as you go

QUICK CHECK

1. What does a sensor do?

A Stores files
B Moves a motor
C Measures something in the physical world and turns it into data
D Displays a picture

2. What does an actuator do?

A Makes something happen in the physical world
B Measures temperature
C Stores data
D Connects to Wi-Fi

3. Which is an actuator?

A A light sensor
B A motor
C A temperature sensor
D A GPS receiver

4. What is a feedback loop?

A A type of cable
B A broken circuit
C A looping video
D When a system's output affects its next input so it can adjust

QUICK CHECK: ANSWERS

1. **C: Measures something in the physical world and turns it into data**
   Sensors are inputs.

2. **A: Makes something happen in the physical world**
   Actuators are outputs.

3. **B: A motor**
   Motors cause movement.

4. **D: When a system's output affects its next input so it can adjust**
   Feedback keeps conditions steady.

3 MARKS · IN YOUR WORKBOOK

EXIT TICKET

Describe how an automatic streetlight works as an input-process-output system.

Look back at today's success criteria: can you tick them all?

- ✓ I can name common sensors and what they measure
- ✓ I can name common actuators and what they do
- ✓ I can describe an IoT system as input, process and output with feedback

LESSON 3 OF 6

SMART HOMES

Smart home devices can save energy, add convenience and help people live independently, but they also collect personal data, can be hacked, may stop working if the company closes down, and can be misused to watch or control others.

STARTER · 5 MINUTES

DO NOW

Would you like a home where the lights, heating and locks all respond to your voice? What could go wrong?

TODAY

LEARNING INTENTIONS

WE ARE LEARNING TO

Evaluate the benefits and risks of smart home devices

SUCCESS CRITERIA: I CAN...

- ✓ I can describe benefits of smart home devices
- ✓ I can describe privacy and security risks
- ✓ I can make a reasoned judgement about a smart home device

BENEFITS

- Convenience: control lights, heating and music by voice or app
- Energy saving: heating only when needed, smart plugs, smart meters
- Independence: reminders and alerts help older or disabled people live at home
- Security: cameras and smart locks you can check from anywhere

KEY WORDS

SMART HOME	A home with devices that can be controlled remotely or automatically.
VOICE ASSISTANT	Software that responds to spoken commands, such as Alexa, Google Assistant or Siri.
PRIVACY	Having control over your personal information and who can see it.
INTEROPERABILITY	The ability of devices or systems from different makers to work together.

RISKS

- Privacy: devices collect data about your routines, voice and video
- Security: weak devices can be hacked and used to spy or attack
- Dependence: if the company's servers shut down, some devices stop working
- Misuse: devices can be used to monitor or control people in the home

MAKING IT WORK TOGETHER

- Interoperability: devices from different brands working together
- New standards like Matter aim to make smart home devices compatible
- Local control means some things keep working without the internet
- Check before you buy: updates, privacy settings, data storage

ACTIVITY · 20 MINUTES · PAIRS

SMART HOME AUDIT

1. Your teacher gives you the privacy policies (simplified) for three smart devices: a speaker, a doorbell camera and a thermostat.
2. For each, record what data it collects, where it goes and how long it's kept.
3. Score each device out of 5 for privacy and explain your score.
4. Write advice for a family deciding whether to buy one.

In your workbook, answer:

? Which device scored lowest for privacy, and why?

? Write your advice to the family.

Your workbook: Lesson 3 · tick each step off as you go

QUICK CHECK

1. Which is a benefit of smart home devices?

A Needing the internet to work
B Collecting data about your routine
C Being hacked
D Saving energy by heating only when needed

2. Which is a privacy risk of smart speakers?

A They use batteries
B They are too loud
C They can't play music
D They may record and send voice data to company servers

3. What is interoperability?

A A type of sensor
B Operating a device with your voice
C Fixing a broken device
D Devices from different makers working together

4. Why might a smart device stop working one day?

A It runs out of data
B The Wi-Fi password is too long
C The company might shut down its servers
D It gets too smart

QUICK CHECK: ANSWERS

1. **D: Saving energy by heating only when needed**
   Smarter heating saves energy.

2. **D: They may record and send voice data to company servers**
   Voice data is personal.

3. **D: Devices from different makers working together**
   Standards like Matter help this.

4. **C: The company might shut down its servers**
   Cloud dependence is a risk.

2 MARKS · IN YOUR WORKBOOK

EXIT TICKET

Describe one benefit and one risk of using a smart doorbell.

Look back at today's success criteria: can you tick them all?

- ✓ I can describe benefits of smart home devices
- ✓ I can describe privacy and security risks
- ✓ I can make a reasoned judgement about a smart home device

LESSON 4 OF 6

SMART CITIES

Smart cities use sensors and data to manage traffic, energy, waste, air quality and public services more efficiently.

STARTER · 5 MINUTES

DO NOW

Your town has 500 bins. Some fill up in a day, others take a week. How could a council empty them more efficiently?

TODAY

LEARNING INTENTIONS

WE ARE LEARNING TO

Describe how smart cities use IoT data to improve services, and discuss the concerns

SUCCESS CRITERIA: I CAN...

- ✓ I can describe examples of smart city technology
- ✓ I can explain how data helps cities make decisions
- ✓ I can discuss concerns about smart cities

DATA-DRIVEN CITIES

- A smart city uses sensors and data to run services better
- Traffic lights that adapt to queues
- Bins that report when they're full, so lorries only collect full ones
- Streetlights that dim when nobody's around

KEY WORDS

SMART CITY	A city that uses sensors, networks and data to run services more efficiently.
OPEN DATA	Data that anyone is allowed to use for free.
SURVEILLANCE	Watching or tracking people, for example with cameras or sensors.
DIGITAL TWIN	A virtual model of a real place or object, kept updated with live data.

BIG IDEAS

- Air-quality sensors show where pollution is worst
- Real-time bus tracking on apps and stops
- Smart parking: find a free space without circling
- Digital twins: virtual copies of a city to test changes before building them

CONCERNS

- Surveillance: cameras and sensors everywhere can track people
- Who owns and controls the data: the council or private companies?
- Cost, maintenance and hacking risks
- Will everyone benefit, or only people with smartphones?

ACTIVITY · 30 MINUTES · GROUPS OF 4

SMART ENNISKILLEN

1. Your group is the council's smart town team for Enniskillen (or your own town).
2. Choose two problems to solve (e.g. traffic on the bridges, flooding from the lough, litter, tourist parking).
3. For each, describe the sensors, data and actions your smart solution would use.
4. Complete a concerns checklist for each solution and present your plan.

In your workbook, answer:

? Describe one smart solution: sensors, data and actions.

? What is the biggest concern about your plan and how would you address it?

Your workbook: Lesson 4 · tick each step off as you go

QUICK CHECK

1. What is a smart city?

A A city that uses sensors and data to run services better
B A city with lots of phone shops
C A very clever city council
D A city with no cars

2. How do smart bins save money?

A They cost nothing
B They empty themselves
C They sort recycling by magic
D Lorries only collect bins that are full

3. What is a digital twin?

A A twin brother online
B A virtual model of a real place, updated with live data
C Two identical phones
D A backup file

4. Which is a concern about smart cities?

A Less traffic
B Surveillance of people
C Cleaner air
D Shorter bus waits

QUICK CHECK: ANSWERS

1. **A: A city that uses sensors and data to run services better**
   Data-driven services.

2. **D: Lorries only collect bins that are full**
   Fewer wasted trips.

3. **B: A virtual model of a real place, updated with live data**
   Used to test changes safely.

4. **B: Surveillance of people**
   Privacy matters.

3 MARKS · IN YOUR WORKBOOK

EXIT TICKET

Describe how sensors could be used to reduce traffic congestion in a town.

Look back at today's success criteria: can you tick them all?

- ✓ I can describe examples of smart city technology
- ✓ I can explain how data helps cities make decisions
- ✓ I can discuss concerns about smart cities

LESSON 5 OF 6

HACKING THE THINGS

Many IoT devices have been sold with weak default passwords and no updates, so attackers can take them over in large numbers to form botnets.

STARTER · 5 MINUTES

DO NOW

A webcam is sold with the username "admin" and password "admin". Why is that a problem when millions are sold?

TODAY

LEARNING INTENTIONS

WE ARE LEARNING TO

Explain why IoT devices are often insecure and describe how laws and users can make them safer

SUCCESS CRITERIA: I CAN...

- ✓ I can explain why IoT devices are often insecure
- ✓ I can describe a botnet attack
- ✓ I can describe the UK's PSTI rules and good security habits

WHY IOT IS EASY TO ATTACK

- Many devices shipped with the same default password, like admin/admin
- Cheap devices often never get security updates
- Owners forget they're connected and never change settings
- Each device is a computer that attackers can control

KEY WORDS

DEFAULT PASSWORD	The password a device comes with from the factory.
BOTNET	A network of devices infected with malware and controlled by criminals.
FIRMWARE	The software built into a device that controls how it works.
PSTI	The UK's Product Security and Telecommunications Infrastructure rules, which set minimum security requirements for smart devices.

BOTNETS

- A botnet is a network of hijacked devices controlled by criminals
- In 2016 the Mirai botnet took over hundreds of thousands of cameras and routers
- It used them to flood an internet company with traffic, taking major websites offline
- Owners usually have no idea their device is involved

THE LAW AND YOU

- The UK PSTI rules (from April 2024) ban universal default passwords on new smart devices
- Makers must say how long they'll provide security updates
- Makers must give a way to report security problems
- You: change passwords, update firmware, use a separate guest network

ACTIVITY · 20 MINUTES · PAIRS

SECURE THE SMART HOME

1. Read the description of the Murphy family's smart home: 8 devices, some old, some with default passwords, all on the main Wi-Fi.
2. Find at least five security problems.
3. Write an action plan, putting the fixes in priority order.
4. Explain which problem the PSTI law would have prevented.

In your workbook, answer:

? List the five problems you found.

? Write your top three fixes and explain the order.

Your workbook: Lesson 5 · tick each step off as you go

QUICK CHECK

1. Why are default passwords dangerous?

A They change every day
B They're too long
C They're the same on many devices and often published online
D They're encrypted

2. What is a botnet?

A A network of robots in a factory
B A network of hijacked devices controlled by criminals
C A chatbot
D A secure network

3. What did the Mirai botnet do in 2016?

A Delivered parcels
B Fixed routers
C Took over IoT devices with default passwords and attacked an internet company
D Made a film

4. Since 2024, what do the UK PSTI rules ban?

A Smart speakers
B Wi-Fi
C All smart devices
D Universal default passwords on new smart devices

QUICK CHECK: ANSWERS

1. **C: They're the same on many devices and often published online**
   Attackers try them first.

2. **B: A network of hijacked devices controlled by criminals**
   Botnets are used for attacks.

3. **C: Took over IoT devices with default passwords and attacked an internet company**
   It caused major website outages.

4. **D: Universal default passwords on new smart devices**
   Each device needs a unique password.

3 MARKS · IN YOUR WORKBOOK

EXIT TICKET

Explain why IoT devices are often less secure than laptops or phones.

Look back at today's success criteria: can you tick them all?

- ✓ I can explain why IoT devices are often insecure
- ✓ I can describe a botnet attack
- ✓ I can describe the UK's PSTI rules and good security habits

LESSON 6 OF 6

DESIGN AN IOT SOLUTION

Good IoT design starts with a real problem and real users, then plans the sensors, actuators, connectivity and data, and builds in security, privacy and sustainability from the start, a principle called "by design".

STARTER · 5 MINUTES

DO NOW

Name one problem at school or in your town that a connected sensor could help with.

TODAY

LEARNING INTENTIONS

WE ARE LEARNING TO

Design an IoT solution to a real problem, considering users, data, security and sustainability

SUCCESS CRITERIA: I CAN...

- ✓ I can define a problem and its users
- ✓ I can plan the hardware, data and rules for an IoT system
- ✓ I can apply security, privacy and sustainability by design

START WITH THE PROBLEM

- Who has the problem, and what do they actually need?
- Would a non-tech solution be simpler?
- What would success look like?
- User stories: "As a ..., I want ..., so that ..."

KEY WORDS

PROTOTYPE	An early working model used to test and improve an idea.
PRIVACY BY DESIGN	Building privacy into a system from the start, for example by collecting only the data needed.
SUSTAINABILITY	Meeting today's needs without harming the environment or future generations.

PLAN THE SYSTEM

- Sensors: what needs measuring?
- Actuators and alerts: what should happen?
- Connectivity and power: Wi-Fi, mobile or low-power network? Battery or mains?
- Data: what's collected, how often, where it goes, who sees it

BUILD IT IN "BY DESIGN"

- Security by design: unique passwords, encryption, updates
- Privacy by design: collect only what's needed; no cameras if a sensor will do
- Sustainability: low power, long-lasting, repairable, recyclable
- Make a prototype and test it with real users

ACTIVITY · 35 MINUTES · GROUPS OF 3

IOT DESIGN CHALLENGE

1. Choose a problem at school or in your town (e.g. classroom windows left open, a car park that's always full, a school garden that dries out).
2. Write a user story and complete the IoT design plan table.
3. Write at least three IF … THEN … rules.
4. Add a "by design" box for security, privacy and sustainability. If you have micro:bits, prototype one part.

In your workbook, answer:

? Write your user story and your three rules.

? Describe how your design is secure, private and sustainable.

Your workbook: Lesson 6 · tick each step off as you go

QUICK CHECK

1. What should IoT design start with?

A The most expensive sensor
B A logo
C A real problem and the users who have it
D The colour scheme

2. What is privacy by design?

A Building privacy in from the start, collecting only the data needed
B Hiding the device
C Using cameras everywhere
D Adding a privacy policy at the end

3. What is a prototype?

A An early working model used to test ideas
B A type of sensor
C A user story
D The final product

4. Which choice makes an IoT design more sustainable?

A Replacing it every year
B Low power use and repairable parts
C Using lots of batteries
D Making it impossible to open

QUICK CHECK: ANSWERS

1. **C: A real problem and the users who have it**
   Start with the problem.

2. **A: Building privacy in from the start, collecting only the data needed**
   Minimise data from day one.

3. **A: An early working model used to test ideas**
   Prototypes help test and improve.

4. **B: Low power use and repairable parts**
   Sustainability means a long, low-impact life.

3 MARKS · IN YOUR WORKBOOK

EXIT TICKET

Explain how privacy by design could be applied to a system that counts people using a library.

Look back at today's success criteria: can you tick them all?

- ✓ I can define a problem and its users
- ✓ I can plan the hardware, data and rules for an IoT system
- ✓ I can apply security, privacy and sustainability by design

END OF TOPIC

INTERNET OF THINGS AND SMART CITIES

Revise each lesson in your workbook: the notes, key words and quick checks. Then take the topic test when Mr Butler sets it.